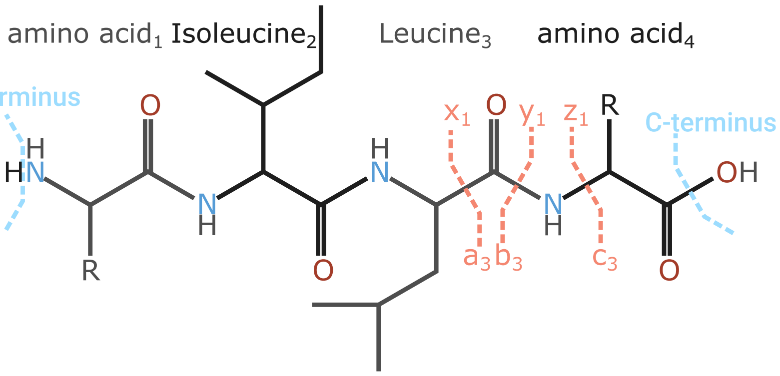
Backbone fragmentation [1,2]



	Formula ($h = H^+ \times z$)	Mass ($h = 1.008 \times z$)
a	$\sum^{n}(AA) - CO + h$	$\sum^{n}(AA) - 27.995 + h$
b	$\sum^{n}(AA) + h$	$\sum^{n}(AA) + h$
c	$\sum^{n}(AA) + NH_3 + h$	$\sum^{n}(AA) + 17.027 + h$
d	$\sum^{n-1}(AA) + C_2H_3N + sc + h$	$\sum^{n-1}(AA) + 41.027 + sc + h$
v	$\sum^{c-1}(AA) + C_2H_2NO + h$	$\sum^{c-1}(AA) + 56.014 + h$
w	$\sum^{c-1}(AA) + C_3H_3O + sc + h$	$\sum^{c-1}(AA) + 55.018 + sc + h$
x	$\sum^{c}(AA) + CO_2 + h$	$\sum^{c}(AA) + 43.990 + h$
y	$\sum^{c}(AA) + H_2O + h$	$\sum^{c}(AA) + 18.011 + h$
z	$\sum^{c}(AA) + O - NH + h$	$\sum^{c}(AA) + 0.984 + h$

sc = sidechain [1,2,3]

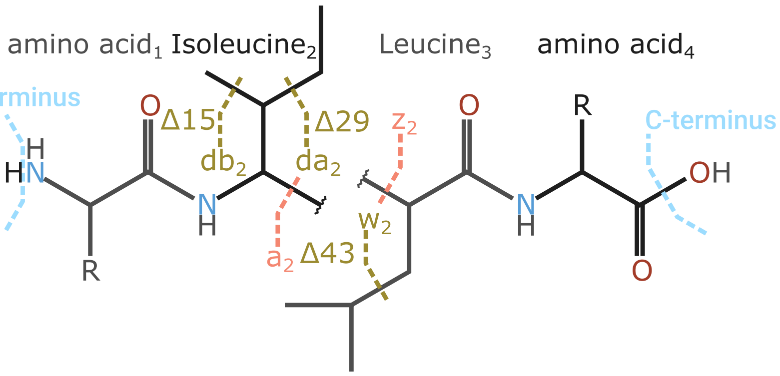
Variant ions

	"/-2H	'/-1H	Base	·/+1H	··/+2H
a			aAU	U	U
b			aAEhU		U
c		e	aAeEU	E	e
d			aAU		
v			aAeU		
w		e	aAeEU	e	
x		U	aAU	U	U
y	U	U	aAeEhU		
z		e	aAeEU	eE	e

Variant ions are ion series differing by one or two hydrogen atoms in mass from the main ion series. The occurance of these is dependent on the fragmentation technique used.

a=EAD
A=EACID
e=ExD[4-8]
E=EtHCD
h=HCD
U=UVPD[9-12]

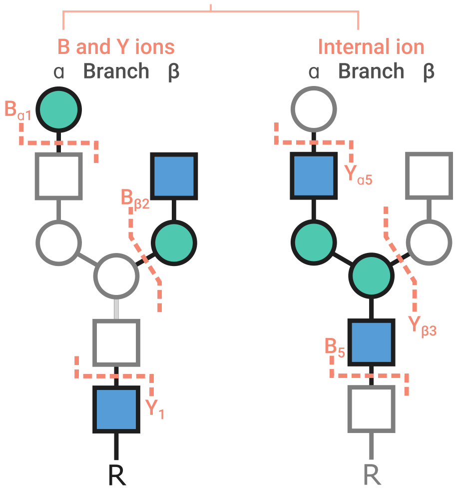
Side chain fragmentation / satellite ions [2,3]



Satellite ions are formed as secondary fragmentation, d from an a ion, w from a z ion. For Threonine and Isoleucine there are two possible d and w ions. These can be listed as 'd/wa$_n$' and 'd/wb$_n$' where a is the heaviest of the two options. The leaving groups for Threonine are CH_3 for wa and OH for wb, for Isoleucine the leaving groups are CH_3 for wa and C_2H_5 for wb. Glycine, alanine, and proline have no satellite ions. The v ion is full loss of the sidechain from a y ion. For ETD and ECD, side chain loss similar to v ions can be observed from many fragments possibly even with multiple side chains lost[4]. Additionally in ECD, w fragmentation can be found in sidechains away from the backbone cleavage[5,6]. Denoted with ^{d}w, where d indicates the distance from the backbone cleavage, '^{0}w', the standard w ion, is written as 'w'.

Glycan [13]

Collectively B and Internal ions are called Oxonium ions when the charge is located on an Oxygen



Ion	Formula	Mass	Ion	Formula	Mass
H-$C_2H_6O_3$	$C_4H_5O_2$	85.0284	S-H_2O	$C_{11}H_{16}NO_7$	274.0921
H-CH_6O_3	$C_5H_5O_2$	97.0284	S	$C_{11}H_{18}NO_8$	292.1027
H-2H_2O	$C_6H_7O_3$	127.0390	G-H_2O	$C_{11}H_{16}NO_8$	290.0870
H-H_2O	$C_6H_9O_4$	145.0495	G	$C_{11}H_{18}NO_9$	308.0976
H	$C_6H_{11}O_5$	163.0601	H+N	$C_{14}H_{24}NO_{10}$	366.1395
N-$C_4H_8O_4$	C_4H_6NO	84.0444	2N	$C_{16}H_{27}N_2O_{10}$	407.1660
N-$C_2H_6O_3$	$C_6H_8NO_2$	126.0550	H+2N	$C_{22}H_{37}N_2O_{15}$	569.2188
N-CH_6O_3	$C_7H_8NO_2$	138.0550	H+N+S	$C_{25}H_{41}N_2O_{18}$	657.2349
N-$C_2H_4O_2$	$C_6H_{10}NO_3$	144.0655			
N-2H_2O	$C_8H_{10}NO_3$	168.0655			
N-H_2O	$C_8H_{12}NO_4$	186.0761			
N	$C_8H_{14}NO_5$	204.0867			

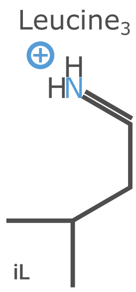
[14]

H=Hex
N=HexNAc
S=Neu5Ac
G=Neu5Gc

Formula & Mass are MH+

Immonium

Formed by double backbone cleavage. Commonly fragments further with neutral losses. These fragments indicate existence of certain amino acids, but not all amino acids readily form immonium ions. The immonium ions have a mass between 0 and 160 Da. Formula: amino acid - CO + H

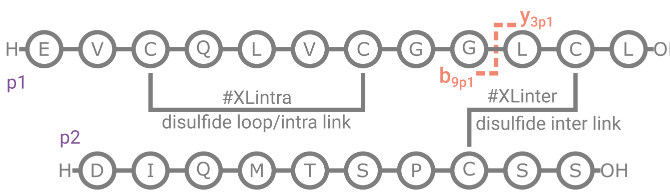


Amino Acid	Formula	Mass	Amino Acid	Formula	Mass
A	C_2H_6N	44.050	K	$C_5H_{13}N_2$	101.108
R	$C_5H_{13}N_4$	129.114	M	$C_4H_{10}NS$	104.053
N	$C_3H_7N_2O$	87.056	F	$C_8H_{10}N$	120.081
D	$C_3H_6NO_2$	88.040	P	C_4H_8N	70.066
C	C_2H_6NS	76.022	S	C_2H_6NO	60.045
E	$C_4H_8NO_2$	102.056	T	C_3H_8NO	74.061
Q	$C_4H_9N_2O$	101.071	W	$C_{10}H_{11}N_2$	159.092
G	CH_4N	30.034	Y	$C_8H_{10}NO$	136.076
H	$C_5H_8N_3$	110.072	V	$C_4H_{10}N$	72.081
I/L/J	$C_5H_{12}N$	86.097			

Formula & Mass are MH+

Cross-links

Note that if you want to see breaking cross-linkers you will need to define your own custom linkers in the Annotator as the breaking is not defined in any of the used modification databases.



Two possible masses for y_{3p1}

y_{3p1} #XLinter LCL + OH - $H_{(broken\ disulfide)}$

y_{3p1} #XLinter LCL + OH - $2H_{(intact\ disulfide)}$ + p2

One possible mass for b_{9p1}

b_{9p1} #XLintra EVCQLVCGG + H - $2H_{(disulfide)}$

This disulfide is a loop link, so it could break either in the linker or in any of the contained amino acids without changing the final mass of the fragment.

To annote these and more on your own data see: github.com/snijderlab/annotator

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V1.2

References
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